Mean Field Control

Bensoussan, Frehse and Yam: Sections 4.1–4.3

Jonatan Wächter

UZH Macro/Finance Reading Group

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Mean field control

- So far, we have mainly studied *mean field games*.
- Many interacting agents, each with a negligible effect on the aggregate: an individual's deviation leaves the mean field unchanged.
- In *mean field control*, we choose a policy for the population and account for how that policy changes the mean field.
- Examples: cooperation among agents; a social planner's problem.

- For a feedback control v and a distribution path m, recall the cost:

$$J(v,m) = \mathbb{E}\left[\int_0^T f\big(x(t), m(t), v(x(t))\big)\,dt + h\big(x(T), m(T)\big)\right]. \tag{2.5}$$

- f is the running cost and h the terminal cost.
- **MFG:** Find an equilibrium pair $(\hat{v}, m)$: $\hat{v}$ is optimal with m held fixed, and m is the state distribution generated by $\hat{v}$.

$$\begin{aligned} J(\hat{v},m) &\le J(v,m), \qquad \forall\, v \text{ admissible},\\ m(t) &= \mathrm{Law}(\hat{x}(t)), \qquad t \in [0,T]. \end{aligned} \tag{2.8}$$

- **MFC:** For each policy v, let m^v be its induced population distribution. Find $\hat{v}$ minimizing $J(v, m^v)$.

$$J(\hat{v}, m^{\hat{v}}) \le J\big(v, m^v\big), \qquad \forall\, v \text{ admissible}. \tag{2.11}$$

Plan for today

- 4.1: Derive HJB-FP optimality system
- 4.2: Other approaches
- 4.3: Stochastic Maximum Principle

4.1 HJB-FP

- have seen in the problem setup how the control v affects the mean field term, indicated by our notation m^v
- To examine properties of optimal control $\hat{v}$ we will employ variational arguments
- Consider $v^\theta = \hat{v} + \theta v$ for some v, and look at $\frac{d}{d\theta} J$

- The running cost f and drift g depend on x, m, v, while the terminal cost h depends on x, m.
- We will get further derivative terms reflecting the effect of variation in v on m^v
- Assume g, f, h differentiable in $m \in L^2$
- Assuming the derivative is a bounded linear functional on L^2, the Riesz representation theorem gives

$$\frac{d}{d\theta} f(x, m + \theta\tilde{m}, v)\Big|_{\theta=0} = \int_{\mathbb{R}^n} \frac{\partial f}{\partial m}(x, m, v)(\xi)\, \tilde{m}(\xi)\, d\xi.$$

- $\frac{\partial f}{\partial m}$ is notation for this L^2 element

- We want to derive properties of an optimal control $\hat{v}$
- Will use adjoint methods, i.e. the idea is to derive a coupled system
  - forward equation for the mean field m
  - backward system for the adjoint u
- and express $\hat{v}$ as $\hat{v} = \hat{v}(x, m, Du(x))$
- This coupled system is (4.12) in BFY, and there's an analogy with (3.11), with the difference being that the adjoint equations contain derivatives w.r.t. m

MFG

$$\begin{aligned} -\partial_t u + Au &= H, \quad & u(x,T) &= h, \\ \partial_t m + A^* m + \operatorname{div}(Gm) &= 0, \quad & m(x,0) &= m_0(x). \end{aligned} \tag{3.11}$$

MFC

$$\begin{aligned} -\partial_t u + Au &= H + \int_{\mathbb{R}^n} \frac{\partial H}{\partial m}(\xi)(x) m(\xi)\, d\xi, \\ u(x,T) &= h + \int_{\mathbb{R}^n} \frac{\partial h}{\partial m}(\xi)(x) m(\xi,T)\, d\xi, \\ \partial_t m + A^* m + \operatorname{div}(Gm) &= 0, \qquad m(x,0) = m_0(x). \end{aligned} \tag{4.12}$$

The MFC problem

- For each feedback v, the state and its density satisfy

$$dx^v = g(x^v, m^v, v(x^v))\,dt + \sigma(x^v)\,dw, \qquad x^v(0) = x_0. \tag{2.9}$$

$$\begin{aligned} \partial_t m^v + A^* m^v + \operatorname{div}(g(x, m^v, v) m^v) &= 0, \\ m^v(x, 0) &= m_0(x). \end{aligned} \tag{4.2}$$

- We minimize the cost over controls v, with m^v determined by the FP equation:

$$\begin{aligned} J(v, m^v) = &\int_0^T \int_{\mathbb{R}^n} f(x, m^v, v)\, m^v \, dx\, dt \\ &+ \int_{\mathbb{R}^n} h(x, m^v(T))\, m^v(x, T)\, dx. \end{aligned} \tag{4.3}$$

A Lagrangian overview

- One can obtain the optimality system from a Lagrangian formulation, with u enforcing the FP constraint.
- Here $J(v, m)$ is the density integral in (4.3), extended to independent v, m, with $m(0) = m_0$ fixed.

$$\Lambda(v, m, u) = J(v, m) - \int_0^T \int_{\mathbb{R}^n} u \left[\partial_t m + A^* m + \operatorname{div}\big(g(x, m, v) m\big) \right] dx\, dt.$$

- Stationarity in u: the FP constraint (4.2).
- Stationarity in m: the adjoint system (4.6).
- Stationarity in v: the first-order control condition (4.7).
- BFY instead differentiate the cost along a control perturbation and introduce an adjoint to eliminate the density variation.

BFY 1/6: Perturb the control

- Start from an optimal feedback $\hat{v}$, with state density $m = m^{\hat{v}}$.
- As in BFY, use v for a perturbation direction and define

$$v^\theta = \hat{v} + \theta v, \qquad \tilde{m}(x,t) := \frac{d}{d\theta} m^{v^\theta}(x,t)\Big|_{\theta=0}.$$

- $\tilde{m}$ is a signed first variation, not a probability density: $\int_{\mathbb{R}^n} \tilde{m}(x,t)\,dx = 0$ for every t.
- Our goal is to compute the resulting change in the actual cost:

$$\delta J := \frac{d}{d\theta} J(v^\theta, m^{v^\theta})\Big|_{\theta=0}.$$

BFY 2/6: Differentiate the FP equation

- Differentiate (4.2) at $\theta = 0$, using:
  - sufficient regularity for exchange in order of differentiation
  - chain rule and product rule (for divergence term)

$$\begin{aligned} \partial_t \tilde{m} + A^* \tilde{m} + \operatorname{div}(g\tilde{m}) + \operatorname{div}\left(\left[\int_{\mathbb{R}^n} \frac{\partial g}{\partial m}(x)(\xi)\tilde{m}(\xi)\, d\xi + \frac{\partial g}{\partial v} v \right] m \right) = 0, \\ \tilde{m}(x,0) = 0. \end{aligned} \tag{4.4}$$

- This is a linear equation for $\tilde{m}$, forced by the control perturbation through $\frac{\partial g}{\partial v} v$.
- Coefficients are evaluated at the reference pair $(\hat{v}, m)$.

BFY 3/6: Differentiate the cost

- The chain rule separates the direct effect of the control from the effect of changing the density.

$$\delta J = \underbrace{\int_0^T \int_{\mathbb{R}^n} f_v \cdot v \, m \, dx \, dt}_{\substack{\text{direct control} \\ \text{contribution}}} + \underbrace{\int_0^T \int_{\mathbb{R}^n} R \, \tilde{m} \, dx \, dt + \int_{\mathbb{R}^n} S \, \tilde{m}(x,T) \, dx}_{\substack{\text{contribution from} \\ \text{changing the density}}}.$$

- Here $f_v = \partial f / \partial v$, and R, S collect the running and terminal cost derivatives with respect to the density:

$$R(x,t) := f + \int_{\mathbb{R}^n} \frac{\partial f}{\partial m}(\xi)(x) m(\xi,t) \, d\xi,$$

$$S(x) := h + \int_{\mathbb{R}^n} \frac{\partial h}{\partial m}(\xi)(x) m(\xi,T) \, d\xi.$$

- The remaining task is to eliminate $\tilde{m}$ from the cost derivative.

BFY 3/6: The full cost derivative (4.5)

- Expanding the schematic gives all five contributions in BFY's cost derivative:

$$\begin{aligned}\delta J = &\int_0^T \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \frac{\partial f}{\partial m}(x)(\xi)\tilde{m}(\xi)m(x)\, d\xi\, dx\, dt \\ &+ \int_0^T \int_{\mathbb{R}^n} f\tilde{m}\, dx\, dt + \int_0^T \int_{\mathbb{R}^n} \frac{\partial f}{\partial v} \cdot v\, m\, dx\, dt \\ &+ \int_{\mathbb{R}^n} h\tilde{m}(x,T)\, dx \\ &+ \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \frac{\partial h}{\partial m}(x)(\xi)\tilde{m}(\xi,T)m(x,T)\, d\xi\, dx.\end{aligned} \tag{4.5}$$

- The density terms account for changes inside f, h and changes in the density used to average costs.

BFY 4/6: Introduce the adjoint

- Choose u so that the terms involving $\tilde{m}$ in (4.5) can be eliminated.
- The backward adjoint equation has running source R and terminal value S:

$$\begin{aligned} -\partial_t u + Au - g \cdot Du - \int_{\mathbb{R}^n} Du(\xi) \cdot \frac{\partial g}{\partial m}(\xi)(x) m(\xi)\, d\xi \\ = f + \int_{\mathbb{R}^n} \frac{\partial f}{\partial m}(\xi)(x) m(\xi)\, d\xi, \\ u(x,T) = h + \int_{\mathbb{R}^n} \frac{\partial h}{\partial m}(\xi)(x) m(\xi,T)\, d\xi. \end{aligned} \tag{4.6}$$

- This is the same adjoint equation obtained by stationarity of the Lagrangian in m.

BFY 5/6: Eliminate the density variation

- Multiply the linearized FP equation (4.4) by u and integrate by parts.
- Use (4.6), $u(T) = S$, and $\tilde{m}(0) = 0$ to obtain

$$\int_0^T \int_{\mathbb{R}^n} R\,\tilde{m}\,dx\,dt + \int_{\mathbb{R}^n} S\,\tilde{m}(x,T)\,dx = \int_0^T \int_{\mathbb{R}^n} Du \cdot \frac{\partial g}{\partial v} v\,m\,dx\,dt.$$

- Substituting into (4.5) leaves a derivative expressed directly in the perturbation v:

$$\delta J = \int_0^T \int_{\mathbb{R}^n} \left[\frac{\partial f}{\partial v} + \left(\frac{\partial g}{\partial v} \right)^* Du \right] \cdot v\,m\,dx\,dt.$$

BFY 6/6: Use optimality

- Since $\hat{v}$ is optimal, $\delta J = 0$ for every admissible perturbation direction v.
- For unconstrained controls, wherever $m > 0$, this gives the first-order condition

$$\frac{\partial f}{\partial v} + \left(\frac{\partial g}{\partial v}\right)^{*} Du = 0. \tag{4.7}$$

- With convexity of $a \mapsto f(x, m, a) + g(x, m, a) \cdot Du$, this identifies a minimizing feedback:

$$\hat{v}(x) = \hat{v}(x, m, Du(x)). \tag{4.8}$$

- Next, use the Hamiltonian to rewrite the FP and adjoint equations as the coupled system (4.12).

- using the definition of the Hamiltonian we get (4.9)

$$f + g \cdot Du = H(x, m, Du). \tag{4.9}$$

- and then differentiate Hamiltonian w.r.t. m and use the envelope theorem to rule out first order contribution from change in control to get (4.10)

$$\int_{\mathbb{R}^n} \left[\frac{\partial f}{\partial m}(\xi)(x) + Du(\xi) \cdot \frac{\partial g}{\partial m}(\xi)(x) \right] m(\xi)\, d\xi = \int_{\mathbb{R}^n} \frac{\partial H}{\partial m}(\xi)(x) m(\xi)\, d\xi. \tag{4.10}$$

- We then insert 4.9 and 4.10 in the equation for the adjoint u (4.6) to get the final system 4.12

$$\begin{aligned} -\partial_t u + Au - g \cdot Du - \int_{\mathbb{R}^n} Du(\xi) \cdot \frac{\partial g}{\partial m}(\xi)(x) m(\xi)\, d\xi \\ = f + \int_{\mathbb{R}^n} \frac{\partial f}{\partial m}(\xi)(x) m(\xi)\, d\xi, \\ u(x,T) = h + \int_{\mathbb{R}^n} \frac{\partial h}{\partial m}(\xi)(x) m(\xi,T)\, d\xi. \end{aligned} \tag{4.6}$$

- We then insert 4.9 and 4.10 in the equation for the adjoint u (4.6) to get the final system 4.12

$$\begin{aligned} -\partial_t u + Au &= H(x, m, Du) + \int_{\mathbb{R}^n} \frac{\partial H}{\partial m}(\xi, m, Du(\xi))(x) m(\xi)\, d\xi, \\ u(x, T) &= h(x, m(T)) + \int_{\mathbb{R}^n} \frac{\partial h}{\partial m}(\xi, m(T))(x) m(\xi, T)\, d\xi. \end{aligned} \tag{4.12}$$

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$$
\begin{aligned}
-\partial_t u + Au &= H(x, m, Du) + \int_{\mathbb{R}^n} \frac{\partial H}{\partial m}(\xi, m, Du(\xi))(x) m(\xi)\, d\xi, \\
u(x, T) &= h(x, m(T)) + \int_{\mathbb{R}^n} \frac{\partial h}{\partial m}(\xi, m(T))(x) m(\xi, T)\, d\xi. \qquad (4.12)
\end{aligned}
$$

$$
\begin{aligned}
\partial_t m + A^* m + \operatorname{div}(G(x, m, Du)m) &= 0, \\
m(x, 0) &= m_0(x).
\end{aligned}
$$

4.2 Other approaches

- another way to represent the cost
- spike variation to drop the convexity assumption from section 4.1

- Will obtain another representation of the cost functional J
- Introduce u^v, where $u^v(x,t)$ is expected cost of starting from state x at time t and running with policy v (until T)

$$\begin{aligned} -\partial_t u^v + Au^v - g(x, m^v, v) \cdot Du^v &= f(x, m^v, v), \\ u^v(x, T) &= h(x, m^v(T)). \end{aligned} \tag{4.14}$$

- Can rewrite J as

$$J(v, m^v) = \int_{\mathbb{R}^n} u^v(x, 0) m_0(x)\, dx. \tag{4.15}$$

- To do this, we put 4.14 into the cost functional, do some integration by parts, use the FP equation and cancel terms

- again want to consider $\frac{d}{d\theta}J$
- This time, differentiate the policy-value function with respect to the perturbation parameter θ:

$$\tilde{u}(x,t) = \frac{d}{d\theta}u^{\hat{v}+\theta v}(x,t)\bigg|_{\theta=0}.$$

- interchanging order of differentiation, chain rule etc gives

$$\begin{aligned} -\partial_t\tilde{u} + A\tilde{u} - g\cdot D\tilde{u} - \left[\int_{\mathbb{R}^n}\frac{\partial g}{\partial m}(x)(\xi)\tilde{m}(\xi)\,d\xi + \frac{\partial g}{\partial v}v\right]\cdot Du^{\hat{v}} \\ = \int_{\mathbb{R}^n}\frac{\partial f}{\partial m}(x)(\xi)\tilde{m}(\xi)\,d\xi + \frac{\partial f}{\partial v}\cdot v, \qquad (4.16) \\ \tilde{u}(x,T) = \int_{\mathbb{R}^n}\frac{\partial h}{\partial m}(x)(\xi)\tilde{m}(\xi,T)\,d\xi. \\ \frac{d}{d\theta}J(\hat{v}+\theta v, m^{\hat{v}+\theta v})\bigg|_{\theta=0} = \int_{\mathbb{R}^n}\tilde{u}(x,0)m_0(x)\,dx. \qquad (4.17) \end{aligned}$$

- BFY claim 4.17 and 4.5 coincide and then we can get the same necessary optimality condition as in section 4.1

Spike variations

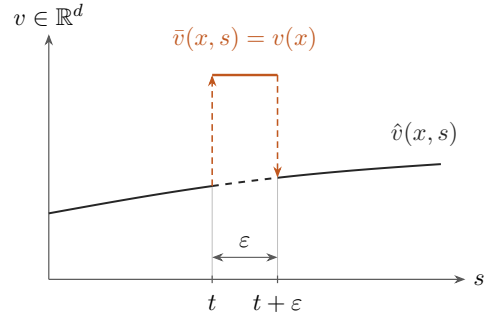
- Recall $L(x, m, v, q) = f(x, m, v) + q \cdot g(x, m, v)$. We want to prove the minimum property, with m and Du held fixed:

$$L(x, m, \hat{v}, Du) \leq L(x, m, v, Du), \qquad \text{for every admissible } v.$$

- for optimal $\hat{v}$, íntroduce $\bar{v}$ with

$$\bar{v}(x, s) = \begin{cases} v(x), & s \in (t, t+\epsilon), \\ \hat{v}(x, s), & s \notin (t, t+\epsilon). \end{cases}$$

- setting $\bar{m} = m_{\bar{v}}$
- $\bar{m}$ will behave just like the density of the optimal control up until t
- after that it follows a different PDE for ϵ time
- remains different after the spike in general



x fixed; schematic for $d > 1$

Rewriting the cost: where L enters

- For any admissible control v, its induced density m^v satisfies the FP equation.
- Subtracting the FP equation multiplied by u leaves the cost unchanged:

$$J(v, m^v) = J(v, m^v) - \int_0^T \int u \underbrace{\left[\partial_s m^v + A^* m^v + \operatorname{div}(g(x, m^v, v) m^v)\right]}_{=0} dx\, ds.$$

- Integration by parts turns the drift term into

$$-\int u \operatorname{div}(g\, m^v)\, dx = \int Du \cdot g\, m^v\, dx.$$

- Together with the running cost f, this gives $L = f + Du \cdot g$.

The rewritten cost—and the comparison

- Integrating the time and diffusion terms by parts gives

$$\begin{aligned} J(v, m^v) = & \int u(x,0) m_0(x)\, dx \\ & + \int_0^T \int \left[L(x, m^v, v, Du) + \partial_s u - Au \right] m^v \, dx \, ds \\ & + \int \left[h(x, m^v(T)) - u(x,T) \right] m^v(x,T) \, dx. \end{aligned}$$

- Apply this identity to $(\bar{v}, \bar{m})$ and $(\hat{v}, m)$, using the same reference adjoint u, and subtract.
- Tricky calculation, a lot of terms cancel

Separate the leading contribution from the remainder

- The leading contribution comes from changing the control during the spike, evaluated at the reference density and adjoint:

$$I_\epsilon = \int_t^{t+\epsilon} \int \left[L(x, m(s), v(x), Du) - L(x, m(s), \hat{v}(x, s), Du) \right] m(x, s)\, dx\, ds.$$

- I_ϵ combines the direct change in running cost with the change in drift valued through Du, which accounts for its first-order consequences for future costs.
- Define R_ϵ as everything left over:

$$\underbrace{J(\bar{v}, \bar{m}) - J(\hat{v}, m)}_{\text{actual cost change}} = \underbrace{I_\epsilon}_{\text{leading contribution}} + \underbrace{R_\epsilon}_{\text{remaining error}}.$$

Let the spike duration tend to zero

- Under suitable regularity and estimates for the perturbed density, one shows

$$\frac{R_\epsilon}{\epsilon} \longrightarrow 0.$$

- Optimality gives $0 \leq I_\epsilon + R_\epsilon$. Dividing by ϵ and taking the limit yields

$$0 \leq \int \big[L(x, m(t), v(x), Du(x,t)) \\ - L(x, m(t), \hat{v}(x,t), Du(x,t))\big] m(x,t)\, dx.$$

- Since the alternative feedback $v(x)$ is arbitrary, $\hat{v}$ must minimize L pointwise:

$$L(x, m(t), \hat{v}(x,t), Du(x,t)) \leq L(x, m(t), v, Du(x,t)),$$
$$\text{for every admissible } v, \quad m(t)\text{-a.e. } x, \quad \text{a.e. } t.$$

4.3 Stochastic Maximum Principle

- recall the HJB-FP system 4.12

$$
\begin{aligned}
-\partial_t u + Au &= H(x, m, Du) + \int_{\mathbb{R}^n} \frac{\partial H}{\partial m}(\xi, m, Du(\xi))(x) m(\xi)\, d\xi, \\
u(x, T) &= h(x, m(T)) + \int_{\mathbb{R}^n} \frac{\partial h}{\partial m}(\xi, m(T))(x) m(\xi, T)\, d\xi. \qquad (4.12) \\
\partial_t m + A^* m + \operatorname{div}(G(x, m, Du)m) &= 0, \\
m(x, 0) &= m_0(x).
\end{aligned}
$$

- goal is to derive stochastic maximum principle from this system

Processes for the SMP

- define X and m

$$X(t) = \hat{x}(t), \qquad m(t) = \mathbb{P}_{X(t)}.$$

- and Y, Z

$$Y(t) = Du(X(t), t), \qquad Z(t) = D^2u(X(t), t)\,\sigma(X(t)),$$
$$V(t) = \hat{v}(X(t), m(t), Y(t)).$$

BSDE for Y

- Apply Itô's formula to $Y(t) = Du(X(t), t)$ and use the differentiated adjoint PDE to obtain the dynamics of Y.
- obtain BSDE for Y given by

$$
\begin{aligned}
-dY(t) = &\left[\frac{\partial f}{\partial x} + \left(\frac{\partial g}{\partial x}\right)^* Y(t) + \operatorname{tr}\left(\left(\frac{\partial \sigma}{\partial x}\right)^* Z(t)\right)\right. \\
&\left. + \mathbb{E}'\left[\frac{\partial^2 f}{\partial x \partial m}(X', m, V')(x) + \left(\frac{\partial^2 g}{\partial x \partial m}(X', m, V')(x)\right)^* Y'\right]\Big|_{x=X(t)}\right] dt \\
&- Z(t)\, dw(t).
\end{aligned}
$$

$$
\begin{aligned}
Y(T) = &\frac{\partial h}{\partial x}(X(T), m(T)) \\
&+ \mathbb{E}'\left[\frac{\partial^2 h}{\partial x \partial m}(X'(T), m(T))(x)\right]\Big|_{x=X(T)}.
\end{aligned}
$$

Notation in the BSDE

$$\frac{\partial^2 f}{\partial x \partial m}(\xi, m, v)(x) = D_x \left[\frac{\partial f}{\partial m}(\xi, m, v)(x) \right].$$

$$\mathbb{E}'[F(X', m, V', Y'; x)]|_{x=X(t)} = \int F(\xi, m, \nu, \eta; x)\, d\mathcal{L}(X, V, Y)(\xi, \nu, \eta) \Big|_{x=X(t)}.$$

- The SMP writes as

$$\mathcal{H}(x,m,v,y) = f(x,m,v) + y \cdot g(x,m,v).$$

$$\begin{aligned}
&X(t), Y(t) \in \mathbb{R}^n, \quad V(t) \in \mathbb{R}^d, \quad Z(t) \in \mathcal{L}(\mathbb{R}^n;\mathbb{R}^n),\\
&dX = g(X,m,V)\,dt + \sigma(X)\,dw,\\
&-dY = \left[\frac{\partial \mathcal{H}}{\partial x}(X,m,V,Y) + \operatorname{tr}\left(\left(\frac{\partial \sigma}{\partial x}(X)\right)^* Z\right)\right.\\
&\qquad \left. + \mathbb{E}'\left[\frac{\partial^2 \mathcal{H}}{\partial x \partial m}(X', m, V', Y')(x)\right]\Big|_{x=X}\right] dt - Z\,dw, \qquad (4.19)\\
&X(0) = x_0,\\
&Y(T) = \frac{\partial h}{\partial x}(X(T), m(T)) + \mathbb{E}'\left[\frac{\partial^2 h}{\partial x \partial m}(X'(T), m(T))(x)\right]\Big|_{x=X(T)}.
\end{aligned}$$

$$V(t) \in \arg\min_v \mathcal{H}(X(t), m(t), v, Y(t)). \qquad (4.20)$$